Ananya Deepak Deoghare

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Summary

Senior Associate Algorithm Engineer — Ranking & Scalable Systems Driving multi-million dollar revenue impact through production ML and optimization research.

Technical Skills

- **Languages:** Python, C++, SQL
- **ML Systems:** Ranking Systems, Feature Stores, A/B Experimentation, Model Serving Pipelines, Optimization Algorithms
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **Infrastructure:** AWS, Hive, Redshift

Experience

Algorithm Engineer Jun 2023 – Present
QuinStreet, Foster City, CA

- Ownership & Business Impact
  - Sole ML owner for the Personal Loans vertical; scope expanded in 2026 to include the conversion model alongside CTR models.
  - Drove 5–8% margin increase by replacing greedy ranking heuristics with optimal assignment (Hungarian Algorithm), improving edge-case handling and advertiser allocation efficiency.
  - Led experimentation roadmap and A/B testing strategy, delivering a projected 10% lift in click-through rate (CTR).
- Modeling & Optimization
  - Designed and productionized Gradient Boosting ranking models with rigorous hyperparameter tuning and cross-validation, improving model accuracy by 5% and reducing error rates by 10%.
  - Built and productionized a conversion prediction model for the Personal Loans funnel, complementing CTR models to optimize downstream conversion.
  - Conducted systematic feature engineering to improve generalization, interpretability, and business metrics.
- Feature Store & ML Infrastructure
  - Owned feature store design and governance for the vertical, ensuring consistency between offline training and online inference.
  - Built automated model validation and evaluation pipelines aligned with revenue KPIs.
  - Partnered with engineering and product to deploy SEO-driven ML systems to production for real-time decision-making.
- Systems & Performance: Own the team's Cython ranking simulator used to stress-test ranking strategies; optimized runtime by 50%, enabling large-scale what-if revenue analysis.

Machine Learning Intern Jun 2022 – Sep 2022
VidMob, New York, NY

- Built deep learning models for creative performance scoring, improving prediction accuracy by 25%.
- Integrated recommendation systems into production ad-tech pipelines.
- Designed model evaluation workflows improving engagement forecasting by 5%.

Student Researcher Jul 2020 – Jan 2023
University of California, Los Angeles (UCLA)

- Co-authored peer-reviewed publication (SIGGRAPH 2022) on multimodal radar + RGB sensing for equitable remote photoplethysmography.
- Proposed shift-robust loss function reducing pulse prediction error by 40%.
- Built bias-mitigation ML pipelines improving fairness and accuracy by 75%.

Education

University of California, Los Angeles (UCLA) 2021 – 2023
M.S. Electrical and Computer Engineering GPA: 3.97/4.0

PES University 2015 – 2019
B.Tech. Electronics and Communication Engineering

Publication

- **SIGGRAPH 2022** — Blending camera and 77 GHz radar sensing for equitable, robust plethysmography